José Cotrim

Automation & Software Engineer

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PROFESSIONAL OVERVIEW

Software and automation engineer with 2+ years of experience in industrial environments, working on both sides of the production line: programming PLCs (Beckhoff TwinCAT, Siemens) and building .NET applications (C#, WPF, SQL Server), machine vision and integration for automotive production lines. Skilled in Git, Azure DevOps and CI/CD.

WORK EXPERIENCE

AUTOMATION & SOFTWARE ENGINEER | Preh, Trofa NOV 2025 – PRESENT

- Led the modernization of legacy industrial desktop applications to a web-based architecture (Angular, ASP.NET Core, Entity Framework Core), keeping integration with machine vision, traceability databases and production equipment.
- Developed Beckhoff TwinCAT 3 applications and integrated industrial equipment through CAN, Ethernet and MQTT, bridging PLCs, sensors and higher-level software systems.
- Developed C#/.NET (WPF/MVVM) applications for manufacturing traceability and configurable production-line parameterization, leveraging SQL Server, stored procedures and JSON-based configurations to enable production changes without recompilation.
- Built machine vision applications with MVTec Halcon, .NET WinUI, Entity Framework and SQL Server (defect detection, brightness analysis, symbol verification), with reusable Halcon libraries packaged as NuGet and deployed as Windows services.
- Migrated safety architectures from ABB Pluto to Beckhoff TwinSAFE (safety I/O, safety relays, electrical schematics, machine validation).
- Developed, commissioned and validated automated production lines and stations for clients such as Mercedes-Benz, BMW, Ford, integrating machine vision, PLCs, test equipment and traceability databases.

PRODUCT DEVELOPMENT ENGINEER | INEGI, Porto SEP 2024 – OCT 2025

- Developed a real-time ergonomic posture analysis system in Python (ZED 2i, ZED SDK body tracking), assessing trunk, head and arm posture and sitting/standing state from 3D keypoints.
- Automated PDF/JSON reports with posture statistics and charts, plus an anonymization algorithm to protect workers' privacy.
- Developed a Universal Robots UR5 collaborative robot program demonstrating human–robot collaboration.

EDUCATION

MSc in Automation Engineering | Faculty of Engineering of University of Porto (FEUP), Porto Graduated: 2024
Erasmus Semester in Technical University of Vienna (TU Wien) Winter Semester 2022
BSc in Mechanical Engineering | Faculty of Engineering of University of Porto (FEUP), Porto Graduated: 2022

SKILLS

C#, .NET, TwinCAT 3, Codesys, ASP.NET Core, Angular, Docker, MVTec Halcon, TwinSAFE, WPF, SQL Server, REST APIs, MVVM, Entity Framework, Azure DevOps, Python, ST/FBD/Ladder, LabVIEW, Siemens TIA Portal, MQTT

LANGUAGES - **Portuguese** Native, **English** C1, **German** A2